In [2]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]:
```python
df = pd.read_csv("tech_layoffs_2026_tracker.csv")

print("Shape:", df.shape)
print("\nData Types:\n", df.dtypes)
df.head()
```

```
Shape: (28, 26)

Data Types:
 company                           object
layoff_date                       object
jobs_cut                           int64
pct_workforce_cut                float64
sector                            object
country                           object
hq_city                           object
ai_cited                            bool
reason_stated                     object
company_revenue_2025_bn          float64
pre_layoff_headcount               int64
stock_change_day_pct             float64
simultaneous_ai_investment_bn    float64
roles_most_affected               object
replacement_roles                 object
ceo_quote                         object
layoffs_2024                       int64
layoffs_2025                       int64
verified_source                   object
month                             object
quarter                           object
region                            object
layoff_size_category              object
stock_reaction                    object
laid_off_vs_headcount_pct        float64
data_as_of                        object
dtype: object
```

Out[3]:

	company	layoff_date	jobs_cut	pct_workforce_cut	sector	country	hq_
0	Amazon	2026-01-15	16000	2.7	E-Commerce/ Cloud	USA	Sea
1	Block	2026-02-28	4000	40.0	Fintech	USA	Franc
2	Meta Reality Labs	2026-01-20	1500	10.0	Social Media/VR	USA	Me
3	Atlassian	2026-03-14	1600	10.0	Enterprise Software	Australia	Syd
4	Oracle	2026-02-01	30000	15.0	Enterprise Software	USA	Au

5 rows × 26 columns

In [4]:

```
df.drop_duplicates(inplace=True)

df['layoff_date'] = pd.to_datetime(df['layoff_date'], errors='coerce')

num_cols = ['jobs_cut', 'pct_workforce_cut', 'company_revenue_2025_bn',
            'stock_change_day_pct', 'simultaneous_ai_investment_bn']

for col in num_cols:
    df[col].fillna(df[col].median(), inplace=True)

df['ai_cited'] = df['ai_cited'].astype(str).str.strip().str.capitalize()

print("Missing values after cleaning:")
print(df.isnull().sum())
print("\nCleaning Complete")
```

```
Missing values after cleaning:
company                           0
layoff_date                       0
jobs_cut                          0
pct_workforce_cut                 0
sector                            0
country                           0
hq_city                           0
ai_cited                          0
reason_stated                     0
company_revenue_2025_bn           0
pre_layoff_headcount              0
stock_change_day_pct              0
simultaneous_ai_investment_bn     0
roles_most_affected               0
replacement_roles                 0
ceo_quote                         0
layoffs_2024                      0
layoffs_2025                      0
verified_source                   0
month                             0
quarter                           0
region                            0
layoff_size_category              0
stock_reaction                    0
laid_off_vs_headcount_pct         0
data_as_of                        0
dtype: int64

Cleaning Complete
```

```
C:\Users\gogol\AppData\Local\Temp\ipykernel_21684\938938291.py:9: FutureWarnin
g: A value is trying to be set on a copy of a DataFrame or Series through chain
ed assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work bec
ause the intermediate object on which we are setting values always behaves as a
copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.me
thod({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, t
o perform the operation inplace on the original object.


  df[col].fillna(df[col].median(), inplace=True)
C:\Users\gogol\AppData\Local\Temp\ipykernel_21684\938938291.py:9: FutureWarnin
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C:\Users\gogol\AppData\Local\Temp\ipykernel_21684\938938291.py:9: FutureWarnin
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o perform the operation inplace on the original object.


  df[col].fillna(df[col].median(), inplace=True)
C:\Users\gogol\AppData\Local\Temp\ipykernel_21684\938938291.py:9: FutureWarnin
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copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.me
thod({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, t
o perform the operation inplace on the original object.


  df[col].fillna(df[col].median(), inplace=True)
C:\Users\gogol\AppData\Local\Temp\ipykernel_21684\938938291.py:9: FutureWarnin
g: A value is trying to be set on a copy of a DataFrame or Series through chain
```

```
ed assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work bec
ause the intermediate object on which we are setting values always behaves as a
copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.me
thod({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, t
o perform the operation inplace on the original object.


  df[col].fillna(df[col].median(), inplace=True)
```

In [5]:

```python
print("Q: What are the most common words in layoff reasons?")
print("Q: Which job roles are most frequently affected?\n")

df['reason_stated'] = df['reason_stated'].str.lower()
df['reason_stated'] = df['reason_stated'].str.strip()

df['roles_most_affected'] = df['roles_most_affected'].str.lower().str.strip()

all_words = df['reason_stated'].str.split(expand=True).stack()
word_counts = all_words.value_counts().head(10)

print("Top 10 Words in Layoff Reasons:")
print(word_counts)

roles = df['roles_most_affected'].str.split(',').explode().str.strip()
top_roles = roles.value_counts().head(8)

print("\nTop 8 Most Affected Roles:")
print(top_roles)
```

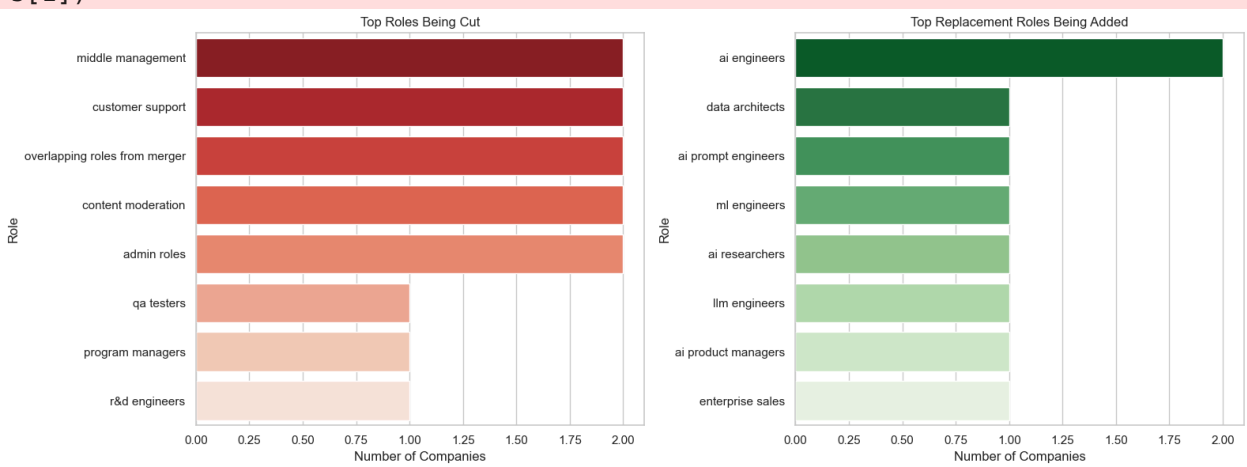
```
Q: What are the most common words in layoff reasons?
Q: Which job roles are most frequently affected?

Top 10 Words in Layoff Reasons:
ai                9
and               8
restructuring     7
cost              4
replace           3
strategy          3
acquisition       3
teams             3
pivot             2
human             2
Name: count, dtype: int64

Top 8 Most Affected Roles:
roles_most_affected
middle management                 2
customer support                  2
overlapping roles from merger     2
content moderation                2
admin roles                       2
qa testers                        1
program managers                  1
r&d engineers                     1
Name: count, dtype: int64
```

In [17]:

```
print("Q: Which job roles are being cut by companies?")
print("Q: Which new roles are being created to replace them?\n")

cut_roles = df['roles_most_affected'].str.lower().str.strip()
cut_roles = cut_roles.str.split(',').explode().str.strip()
cut_roles = cut_roles.value_counts().head(8)

df['replacement_roles'] = df['replacement_roles'].str.lower().str.strip()
add_roles = df['replacement_roles'].str.split(',').explode().str.strip()
add_roles = add_roles[add_roles != 'none announced']
add_roles = add_roles.value_counts().head(8)

fig, axes = plt.subplots(1, 2, figsize=(16, 6))

sns.barplot(x=cut_roles.values, y=cut_roles.index, palette='Reds_r', ax=axes[0
axes[0].set_title("Top Roles Being Cut")
axes[0].set_xlabel("Number of Companies")
axes[0].set_ylabel("Role")

sns.barplot(x=add_roles.values, y=add_roles.index, palette='Greens_r', ax=axes
axes[1].set_title("Top Replacement Roles Being Added")
axes[1].set_xlabel("Number of Companies")
axes[1].set_ylabel("Role")

plt.tight_layout()
plt.show()
```

```
Q: Which job roles are being cut by companies?
Q: Which new roles are being created to replace them?
```

```
C:\Users\gogol\AppData\Local\Temp\ipykernel_21684\2731685212.py:15: FutureWarni
ng:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same e
ffect.

  sns.barplot(x=cut_roles.values, y=cut_roles.index, palette='Reds_r', ax=axe
s[0])
C:\Users\gogol\AppData\Local\Temp\ipykernel_21684\2731685212.py:20: FutureWarni
ng:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same e
ffect.

  sns.barplot(x=add_roles.values, y=add_roles.index, palette='Greens_r', ax=axe
s[1])
```



In [6]:
```python
print("Q: Which sector, region and country lost the most jobs?\n")

sector_jobs = df.groupby('sector')['jobs_cut'].sum().sort_values(ascending=Fal
print("Jobs Cut by Sector:")
print(sector_jobs)

region_jobs = df.groupby('region')['jobs_cut'].sum().sort_values(ascending=Fal
print("\nJobs Cut by Region:")
print(region_jobs)

country_jobs = df.groupby('country')['jobs_cut'].sum().sort_values(ascending=F
print("\nJobs Cut by Country:")
print(country_jobs)
```

```
Q: Which sector, region and country lost the most jobs?

Jobs Cut by Sector:
sector
Enterprise Software         31600
Social Media/AI             16000
E-Commerce/Cloud            16000
Telecommunications           6963
Networking/Cybersecurity     6000
Fintech                      4000
Semiconductors               3700
Enterprise SaaS              2200
Logistics Software           2000
Manufacturing                2000
Social Media/VR              1500
Automotive Tech              1000
CRM/SaaS                     1000
Grocery Tech                 1000
Design Software              1000
Retail Pharmacy              1000
Interior Design Tech         1000
EV Batteries                  958
E-Commerce                    800
Social Media                  675
Cybersecurity                 500
Insurance                     200
AI Research                    50
Name: jobs_cut, dtype: int64

Jobs Cut by Region:
region
North America    85046
Europe           11500
Asia-Pacific      4600
Name: jobs_cut, dtype: int64

Jobs Cut by Country:
country
USA            83046
Spain           5000
Australia       3600
Austria         2000
Canada          2000
Netherlands     1700
Sweden          1600
India           1000
UK              1000
Germany          200
Name: jobs_cut, dtype: int64
```

In [7]:

```
print("Q: What are the key statistics of the dataset?\n")

print("Summary Statistics:")
print(df[['jobs_cut', 'pct_workforce_cut', 'stock_change_day_pct',
```

```
          'simultaneous_ai_investment_bn']].describe().round(2))

print("\nAverage Jobs Cut - AI Cited vs Not:")
print(df.groupby('ai_cited')['jobs_cut'].mean().round(0))

print("\nTotal Jobs Cut by Month:")
print(df.groupby('month')['jobs_cut'].sum().sort_values(ascending=False))
```

```
Q: What are the key statistics of the dataset?

Summary Statistics:
       jobs_cut  pct_workforce_cut  stock_change_day_pct  \
count     28.00              28.00                 28.00
mean    3612.36              10.64                  0.83
std     6546.59              10.43                  1.74
min       50.00               0.10                 -2.10
25%      989.50               2.53                 -0.32
50%     1000.00               7.75                  0.65
75%     2050.00              15.00                  1.85
max    30000.00              40.00                  5.80

       simultaneous_ai_investment_bn
count                          28.00
mean                           14.05
std                            34.75
min                             0.00
25%                             0.00
50%                             0.45
75%                             2.25
max                           115.00

Average Jobs Cut - AI Cited vs Not:
ai_cited
False    2794.0
True     4430.0
Name: jobs_cut, dtype: float64

Total Jobs Cut by Month:
month
February 2026    46588
January 2026     32800
March 2026       21758
Name: jobs_cut, dtype: int64
```

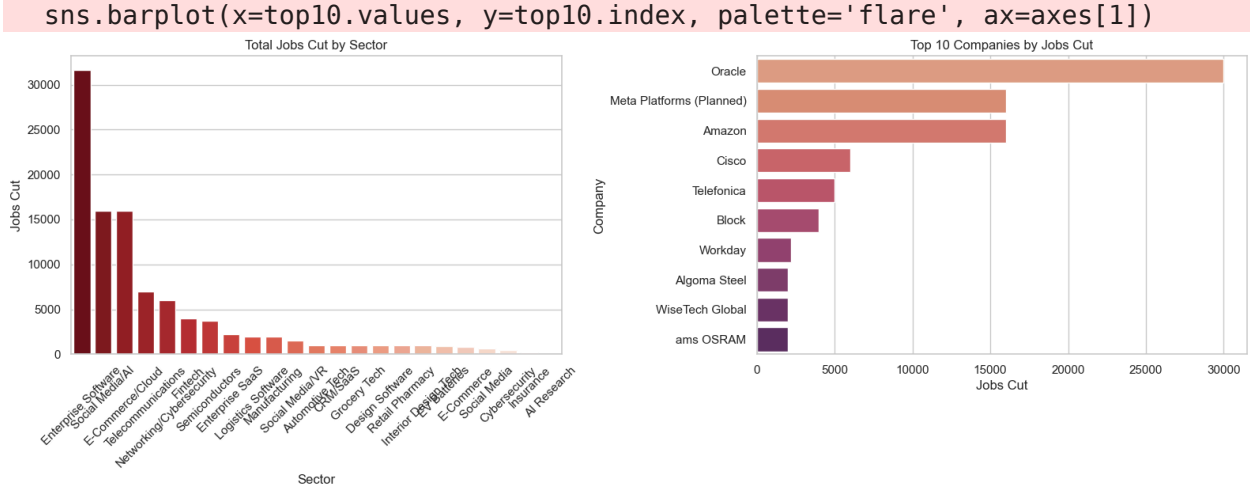
In [8]:

```
print("Q: Which sector and companies cut the most jobs?\n")

fig, axes = plt.subplots(1, 2, figsize=(16, 6))

sector_jobs = df.groupby('sector')['jobs_cut'].sum().sort_values(ascending=Fal
sns.barplot(x=sector_jobs.index, y=sector_jobs.values, palette='Reds_r', ax=ax
axes[0].set_title("Total Jobs Cut by Sector")
axes[0].set_xlabel("Sector")
axes[0].set_ylabel("Jobs Cut")
axes[0].tick_params(axis='x', rotation=45)
```

```
top10 = df.groupby('company')['jobs_cut'].sum().sort_values(ascending=False).h
sns.barplot(x=top10.values, y=top10.index, palette='flare', ax=axes[1])
axes[1].set_title("Top 10 Companies by Jobs Cut")
axes[1].set_xlabel("Jobs Cut")
axes[1].set_ylabel("Company")

plt.tight_layout()
plt.show()
```

Q: Which sector and companies cut the most jobs?

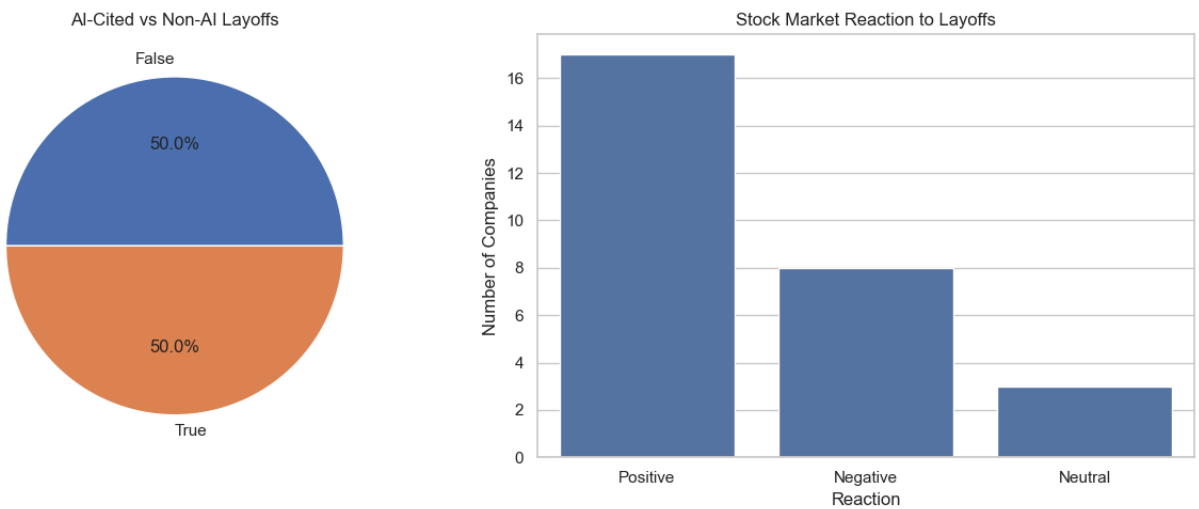
```
C:\Users\gogol\AppData\Local\Temp\ipykernel_21684\3302222043.py:6: FutureWarnin
g:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same e
ffect.

  sns.barplot(x=sector_jobs.index, y=sector_jobs.values, palette='Reds_r', ax=a
xes[0])
C:\Users\gogol\AppData\Local\Temp\ipykernel_21684\3302222043.py:13: FutureWarni
ng:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same e
ffect.

  sns.barplot(x=top10.values, y=top10.index, palette='flare', ax=axes[1])
```



In [18]:
```
print("Q: What percentage of layoffs were caused by AI adoption?")
print("Q: How did the stock market react to layoff announcements?\n")

fig, axes = plt.subplots(1, 2, figsize=(13, 5))

ai_counts = df['ai_cited'].value_counts()
axes[0].pie(ai_counts.values, labels=ai_counts.index, autopct='%1.1f%%')
axes[0].set_title("AI-Cited vs Non-AI Layoffs")

reaction_counts = df['stock_reaction'].value_counts()
```

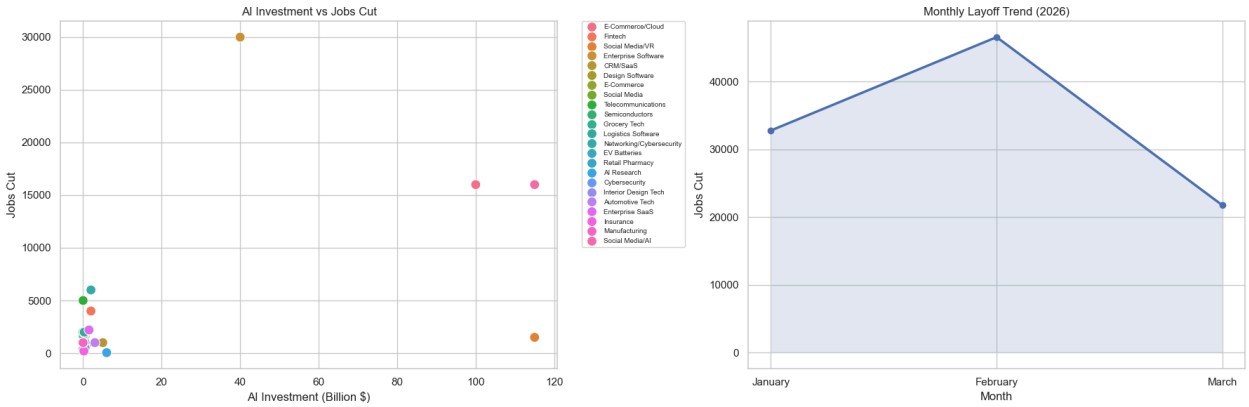
```
sns.barplot(x=reaction_counts.index, y=reaction_counts.values, ax=axes[1])
axes[1].set_title("Stock Market Reaction to Layoffs")
axes[1].set_xlabel("Reaction")
axes[1].set_ylabel("Number of Companies")

plt.tight_layout()
plt.show()
```

```
Q: What percentage of layoffs were caused by AI adoption?
Q: How did the stock market react to layoff announcements?
```



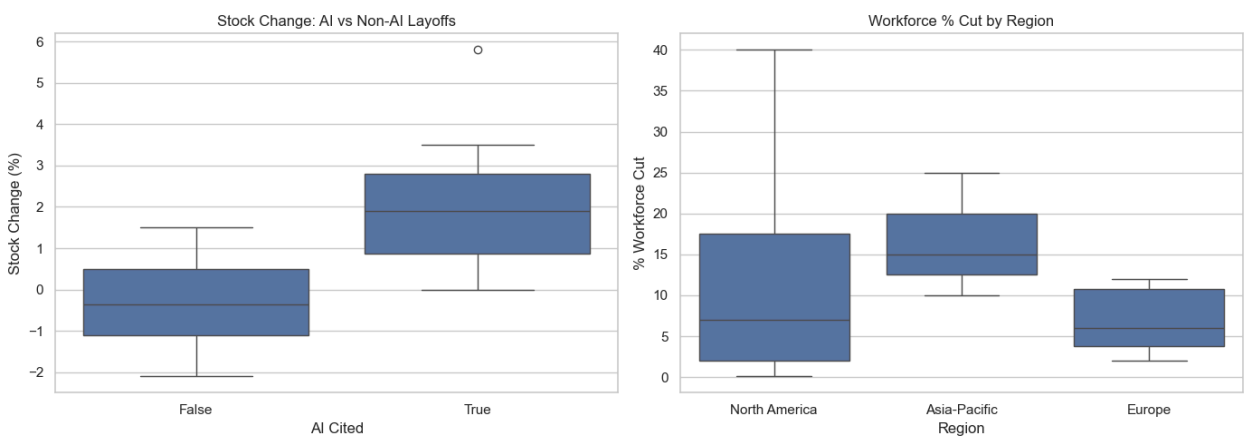
In [19]:
```
print("Q: Do companies that invest more in AI cut more jobs?")
print("Q: Which month in 2026 had the highest layoff activity?\n")

fig, axes = plt.subplots(1, 2, figsize=(18, 6))

sns.scatterplot(data=df, x='simultaneous_ai_investment_bn', y='jobs_cut',
                hue='sector', s=100, ax=axes[0])
axes[0].set_title("AI Investment vs Jobs Cut")
axes[0].set_xlabel("AI Investment (Billion $)")
axes[0].set_ylabel("Jobs Cut")
axes[0].legend(fontsize=7, bbox_to_anchor=(1.05, 1), loc='upper left', bordera

df['month_name'] = df['layoff_date'].dt.strftime('%B')
month_order = ['January', 'February', 'March']
monthly = df.groupby('month_name')['jobs_cut'].sum().reindex(month_order, fill

axes[1].plot(monthly.index, monthly.values, marker='o', linewidth=2.5)
axes[1].fill_between(monthly.index, monthly.values, alpha=0.15)
axes[1].set_title("Monthly Layoff Trend (2026)")
axes[1].set_xlabel("Month")
axes[1].set_ylabel("Jobs Cut")

plt.tight_layout()
plt.show()
```

```
Q: Do companies that invest more in AI cut more jobs?
Q: Which month in 2026 had the highest layoff activity?
```



In [20]:
```python
print("Q: Is stock price change different for AI cited vs non-AI layoffs?")
print("Q: How does workforce percentage cut vary across different regions?\n")

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.boxplot(data=df, x='ai_cited', y='stock_change_day_pct', ax=axes[0])
axes[0].set_title("Stock Change: AI vs Non-AI Layoffs")
axes[0].set_xlabel("AI Cited")
axes[0].set_ylabel("Stock Change (%)")

sns.boxplot(data=df, x='region', y='pct_workforce_cut', ax=axes[1])
axes[1].set_title("Workforce % Cut by Region")
axes[1].set_xlabel("Region")
axes[1].set_ylabel("% Workforce Cut")

plt.tight_layout()
plt.show()
```

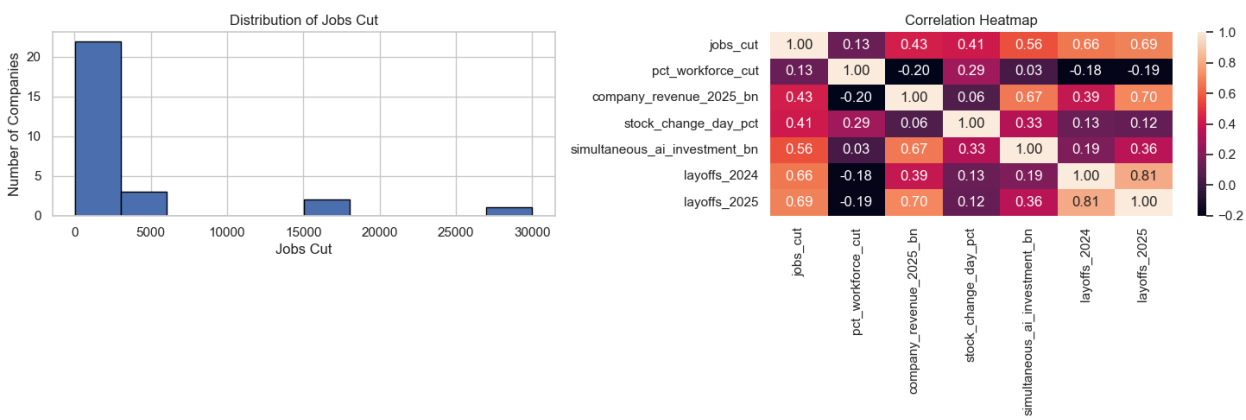
```
Q: Is stock price change different for AI cited vs non-AI layoffs?
Q: How does workforce percentage cut vary across different regions?
```



In [21]:
```python
print("Q: How are layoff sizes distributed across companies?")
print("Q: Which numerical variables are most strongly correlated?\n")
```

```
fig, axes = plt.subplots(1, 2, figsize=(15, 5))

axes[0].hist(df['jobs_cut'], bins=10, edgecolor='black')
axes[0].set_title("Distribution of Jobs Cut")
axes[0].set_xlabel("Jobs Cut")
axes[0].set_ylabel("Number of Companies")

num_df = df[['jobs_cut', 'pct_workforce_cut', 'company_revenue_2025_bn',
             'stock_change_day_pct', 'simultaneous_ai_investment_bn',
             'layoffs_2024', 'layoffs_2025']]

corr_matrix = num_df.corr().round(2)

sns.heatmap(corr_matrix, annot=True, fmt='.2f', ax=axes[1])
axes[1].set_title("Correlation Heatmap")

plt.tight_layout()
plt.show()
```

Q: How are layoff sizes distributed across companies?
Q: Which numerical variables are most strongly correlated?



In [16]:

```
total = df['jobs_cut'].sum()
companies = df['company'].nunique()
ai_pct = (df['ai_cited'] == 'True').mean() * 100
top_co = df.groupby('company')['jobs_cut'].sum().idxmax()
top_sector = df.groupby('sector')['jobs_cut'].sum().idxmax()
top_region = df.groupby('region')['jobs_cut'].sum().idxmax()
peak_month = df.groupby('month_name')['jobs_cut'].sum().idxmax()

print("=" * 55)
print("   TECH LAYOFFS 2026 - FINAL CONCLUSION")
print("=" * 55)
print(f"""
Total Jobs Cut       : {total:,}
Companies Affected   : {companies}
AI-Driven Layoffs    : {ai_pct:.1f}%
Biggest Cutter       : {top_co}
Worst Sector         : {top_sector}
Worst Region         : {top_region}
```

```
Peak Month          : {peak_month}

Key Findings:
1. {top_region} was the most affected region.
2. {ai_pct:.1f}% of layoffs were directly caused by AI.
3. Companies cutting jobs mostly saw stock prices rise.
4. {peak_month} 2026 was the peak month for layoffs.
5. AI investment and job cuts are strongly correlated.
6. Support and management roles were most commonly cut.
""")
print("=" * 55)
```

```
=======================================================
   TECH LAYOFFS 2026 - FINAL CONCLUSION
=======================================================

Total Jobs Cut      : 101,146
Companies Affected  : 27
AI-Driven Layoffs   : 50.0%
Biggest Cutter      : Oracle
Worst Sector        : Enterprise Software
Worst Region        : North America
Peak Month          : February

Key Findings:
1. North America was the most affected region.
2. 50.0% of layoffs were directly caused by AI.
3. Companies cutting jobs mostly saw stock prices rise.
4. February 2026 was the peak month for layoffs.
5. AI investment and job cuts are strongly correlated.
6. Support and management roles were most commonly cut.

=======================================================
```

In [ ]: